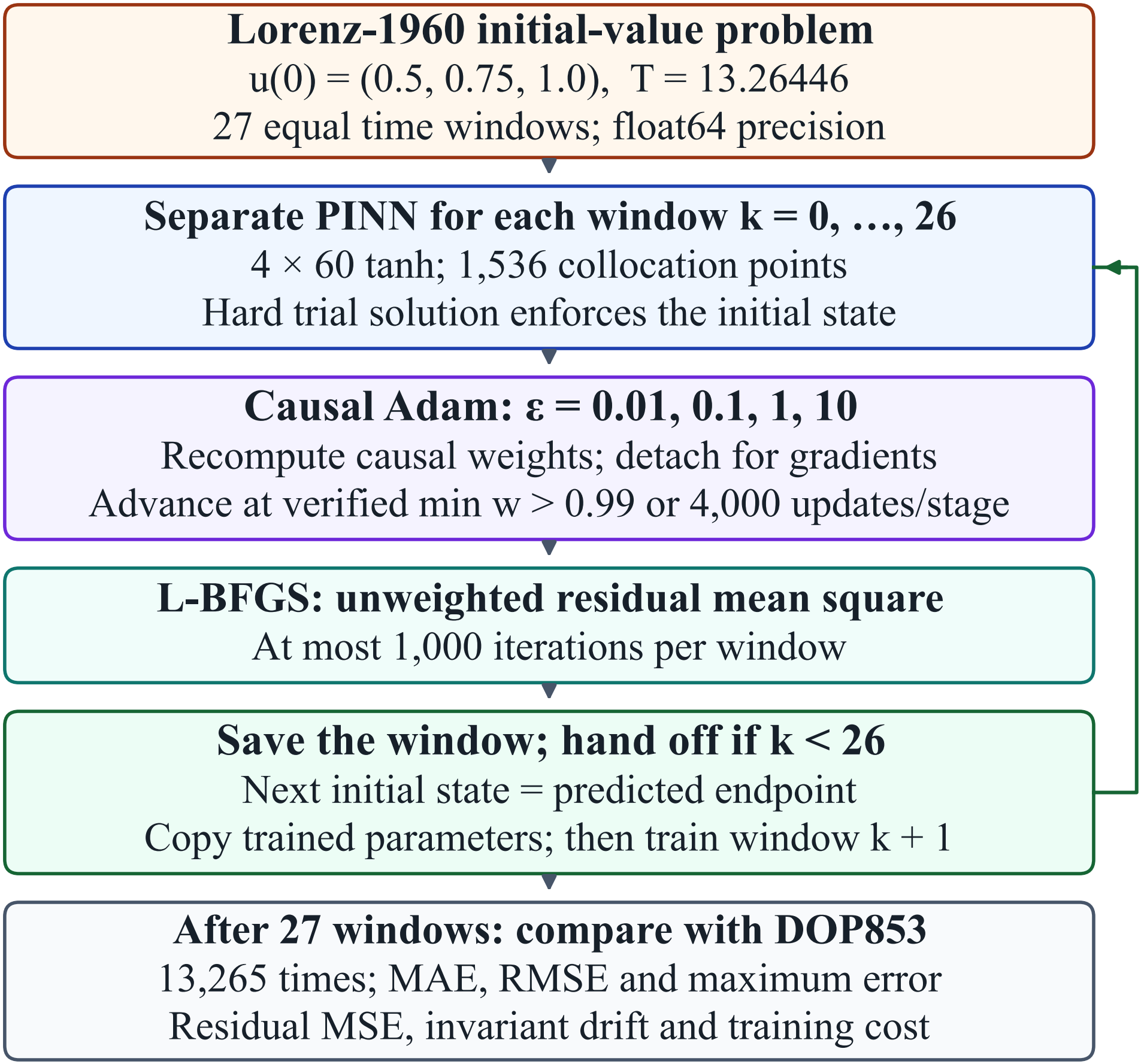

Lorenz-1960 initial-value problem
u(0) = (0.5, 0.75, 1.0), T = 13.26446
27 equal time windows; float64 precision
Separate PINN for each window k = 0, …, 26
4 × 60 tanh; 1,536 collocation points
Hard trial solution enforces the initial state
Causal Adam: ε = 0.01, 0.1, 1, 10
Recompute causal weights; detach for gradients
Advance at verified min w > 0.99 or 4,000 updates/stage
L-BFGS: unweighted residual mean square
At most 1,000 iterations per window
Save the window; hand off if k < 26
Next initial state = predicted endpoint
Copy trained parameters; then train window k + 1
After 27 windows: compare with DOP853
13,265 times; MAE, RMSE and maximum error
Residual MSE, invariant drift and training cost